

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 11

1's COMPLEMENT AND 2's COMPLEMENT

AIM: To write and execute an 8085 assembly language program to find 1's Complement and 2's Complement and store the result in memory.

ALGORITHM FOR 1's COMPLEMENT

1. Load the 8-bit number into accumulator A.
2. Use CMA instruction to complement all bits.
3. Store the result.
4. Stop.

CODE:

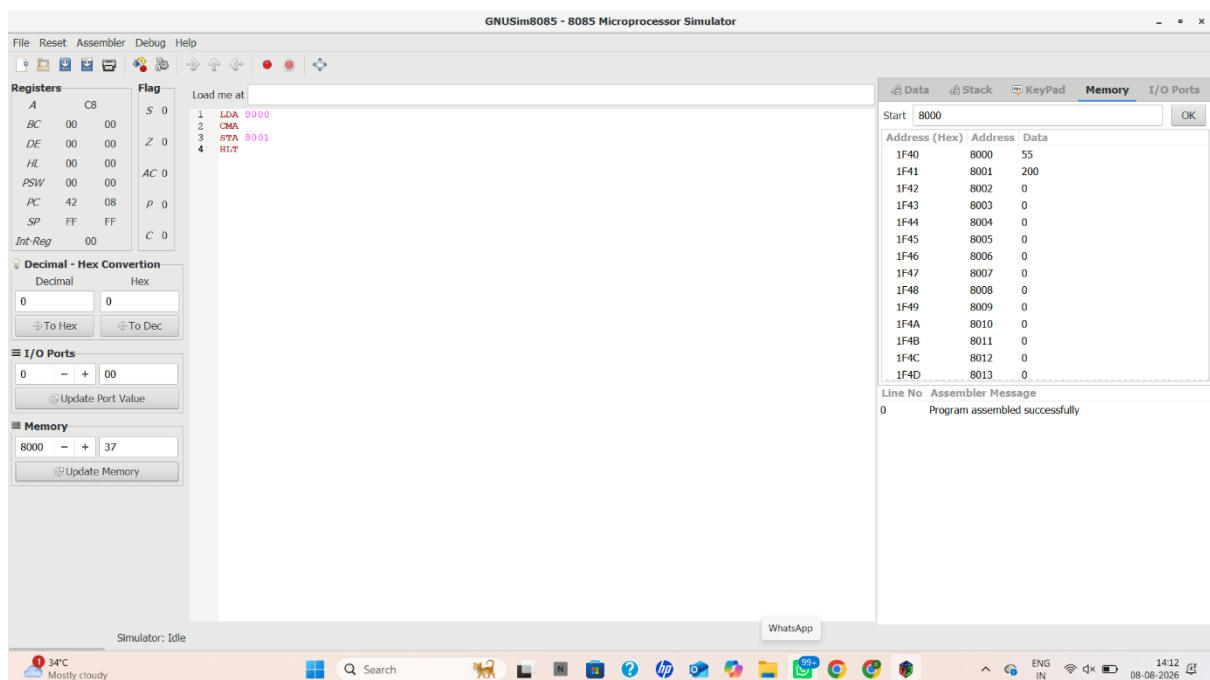
```
MVI A,25H
```

```
CMA
```

```
STA 8050H
```

```
HLT
```

OUTPUT:



ALGORITHM FOR 2's COMPLEMENT:

1. Load the 8-bit number into accumulator A.
2. Use CMA instruction to find 1's complement.
3. Increment A by 1 using INR A.
4. Store the result.
5. Stop.

CODE:

MVI A,25H

CMA

INR A

STA 8050H

HLT

OUTPUT:

